

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Circles	Hard

Question

Point F lies on a unit circle in the xy -plane and has coordinates $(1, 0)$. Point G is the center of the circle and has coordinates $(0, 0)$. Point H also lies on the circle and has coordinates $(-1, y)$, where y is a constant. Which of the following could be the positive measure of angle FGH , in radians?

Answer

- A. $\frac{27\pi}{2}$
- B. $\frac{29\pi}{2}$
- C. 24π
- D. 25π

Correct Answer: D

Rationale

Choice D is correct. It's given that the circle is a unit circle, which means the circle has a radius of 1. It's also given that point G is the center of the circle and has coordinates $(0, 0)$ and that point H lies on the circle and has coordinates $(-1, y)$. Since the radius of the circle is 1, the value of y must be 0, as all other points with an x -coordinate of -1 are a distance greater than 1 away from point G . Since F and H are points on the unit circle centered at G , let side FG be the starting side of the angle and side GH be the terminal side of the angle. Since side FG is on the positive x -axis and side GH is on the negative x -axis, side FG is half of a rotation around the unit circle, or π radians, away from side GH . Therefore, the positive measure of angle FGH , in radians, is equal to π plus an integer multiple of 2π . In other words, the positive measure of angle FGH , in radians, is an odd integer multiple of π . Of the given choices, only 25π is an odd integer multiple of π .

Choice A is incorrect. This could be the positive measure of an angle where the starting side is FG and the terminal side contains the point $(0, -1)$, not $(-1, 0)$.

Choice B is incorrect. This could be the positive measure of an angle where the starting side is FG and the terminal side contains the point $(0, 1)$, not $(-1, 0)$.

Choice C is incorrect. This could be the positive measure of an angle where the starting side is FG and the terminal side contains the point $(1, 0)$, not $(-1, 0)$.